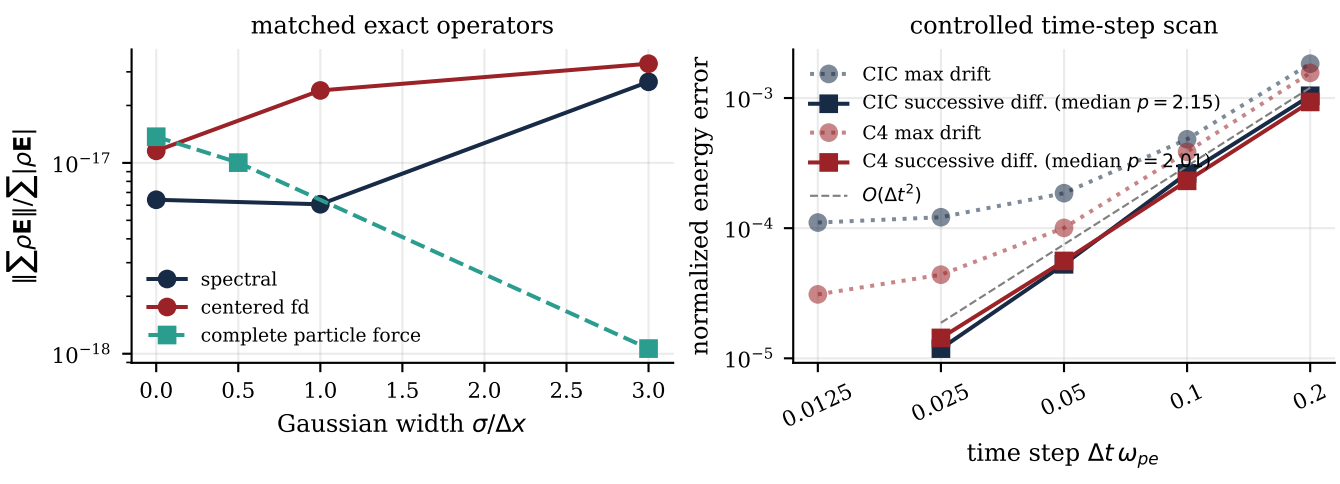
matched exact operators
$\|\sum \rho \mathbf{E}\| / \sum |\rho \mathbf{E}|$
10^{-17}
10^{-18}
spectral
centered fd
complete particle force
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Gaussian width $\sigma/\Delta x$
controlled time-step scan
normalized energy error
10^{-3}
10^{-4}
10^{-5}
CIC max drift
CIC successive diff. (median $p = 2.15$)
C4 max drift
C4 successive diff. (median $p = 2.01$)
$O(\Delta t^2)$
0.0125
0.025
0.05
0.1
0.2
time step $\Delta t\, \omega_{pe}$